Reproducible Terrain-Aware Repeater Field-Strength Mapping with ITU-R P.1812

Beat W. Meier, HB9MQB

19 July 2026

Abstract

Coverage maps for terrestrial repeaters are often rendered as binary or single-colour visibility regions. Such products do not preserve a physical field quantity and cannot distinguish weak from strong predicted service. This paper specifies an open, reproducible workflow that calculates path-specific electric field strength in dB µV m^{-1} using Recommendation ITU-R P.1812-8, terrain profiles over a full 360°, actual effective radiated power (ERP) where available, and an explicit P.525 free-space treatment within the short-path region in which a valid P.1812 profile cannot yet be formed. Polar results are interpolated to an azimuthal-equidistant raster, reprojected, and published both as numerical and colour tiles. A cursor service reports field strength and, separately, an expected nominal S-meter indication under stated ideal receiving assumptions. The implementation uses Apple M5 performance-core scheduling, shared memory, and an optional fused MLX/Metal interpolation kernel while retaining a numerically equivalent NumPy backend. We describe inputs, equations, numerical contracts, validation layers, uncertainty, and portability. The method improves information content and reproducibility over binary terrain-visibility maps; it remains a planning prediction rather than a guarantee of communication. **Keywords:** radio propagation; field strength; ITU-R P.1812; terrain profile; repeater; GIS; S meter; Apple silicon; reproducibility

1 Introduction

Maps are an efficient interface between radio-propagation models and operators, but the graphical form can obscure the modeled quantity. A terrain horizon or binary viewshed answers whether two points have geometric visibility under a particular surface model. It does not calculate received field strength, nor does it represent diffraction, troposcatter, anomalous propagation, frequency, ERP, receiver height, radio-climatic zone, or statistical time and location percentages. A single-colour polygon consequently invites an interpretation that its boundary separates reliable service from no service.

For terrestrial point-to-area applications from 30 MHz to 6 GHz, ITU-R P.1812 provides a path-specific prediction method that combines several propagation mechanisms and returns basic transmission loss and field strength [7]. The method builds upon decades of work on terrestrial propagation and diffraction, including the Bullington and multiple-knife-edge constructions [2, 3].

This work contributes a reproducible computational and publication contract: (i) physically meaningful dB µV m^{-1} rasters, (ii) stable numerical web tiles distinct from presentation colours, (iii) point sampling with an explicitly conditional S-meter estimate, (iv) provenance manifests, and (v) a native Apple M5 execution strategy with documented routes to other operating systems and accelerators. The work does not introduce a new propagation law; its novelty is the auditable integration of a recognized path model, geospatial sampling, numerical web publication, and receiver-reference interaction.

2 Propagation basis

2.1 Path-specific field prediction

Recommendation P.1812-8 predicts terrestrial point-to-area propagation from a terrain profile and terminal geometry. Its valid frequency interval is 30–6000 MHz; paths use ground height, representative clutter height, radio-climatic zones, antenna heights above local ground, polarization, coordinates, and statistical percentages [7]. The selected open implementation is Py1812 at an exact revision rather than an unpinned package release [9].

For every receiver endpoint, the implementation passes frequency f, time percentage p, profile distance d_i, terrain h_i, representative clutter R_i, climatic zone z_i, transmitter and receiver heights h_{tg} and h_{rg}, polarization, terminal coordinates, location percentage p_L, and ERP to P.1812. The model internally accounts for the effective Earth geometry, diffraction, anomalous/-ducting mechanisms and troposcatter. The wrapper must not add a second Earth-curvature correction.

2.2 Near-zone treatment

P.1812 requires an adequate terrain profile; the selected implementation needs at least five profile points and a path of approximately 0.25 km. Inside this region we use the free-space relation associated with P.525 [6]. Expressed for ERP in watts and distance in kilometres,

$$E_{\mathrm{fs}} = 76.92 + 10\log_{10} P_{\mathrm{ERP}} - 20\log_{10} d \quad [\mathrm{dB}\mu\mathrm{V/m}]. \tag{1}$$

This branch is restricted to the short-path domain. It is not a substitute for P.1812 across the service area.

The distinction matters in obstructed terrain, where free-space extension would systematically overstate distant field.

2.3 Transmitter and receiver assumptions

The calculation uses the reported ERP whenever present; 12 W is an explicit fallback only for missing values. ERP must not be confused with transmitter output power before feed-line and antenna effects. The radius is capped at 100 km, receiver height defaults to 1.5 m above ground, and the default time/location percentages are 50/50. Vertical polarization is the default but is a station input.

The present reference implementation applies zero representative clutter height unless a future, provenance-bearing clutter adapter is selected. This limitation is recorded in every manifest. An unrecorded assumption would be more harmful than a conservative, visible simplification.

3 Geospatial method

3.1 Terrain inputs and vertical meaning

The engine accepts georeferenced elevation rasters and transforms a bounded domain into a local azimuthal-equidistant (AEQD) coordinate system centred on the transmitter. AEQD preserves distance and direction from the centre, which matches radial path extraction.

Copernicus DEM GLO-30 is a useful worldwide default, but it is a digital surface model containing vegetation and infrastructure rather than a universal bare-earth DTM. It has one-arc-second latitude spacing and EGM2008 vertical heights; reported global aggregate accuracies allow local deviations [1]. Comparative research also demonstrates that DEM errors depend on terrain and product type [8]. Therefore, input post spacing must not be equated with propagation accuracy. Regional DTMs may be substituted when their datum, license, version, and preprocessing are retained.

3.2 Polar sampling

Let R be calculation radius and Δ_a the requested maximum outer-arc spacing. With a minimum of 360 rays, the ray count is

$$N_\theta = \max\left(360, \left\lceil \frac{2\pi R}{\Delta_a} \right\rceil\right). \tag{2}$$

Azimuths are uniformly distributed in $[0, 360)$, avoiding duplication of the zero-degree ray. Terrain defaults to a 50 m profile step. Field endpoints default to 200 m radial spacing. For every azimuth, the terrain prefix from transmitter to each endpoint is evaluated by P.1812. This is computationally expensive but retains the physical endpoint dependence.

Where an externally prepared climatic-zone raster is present, categorical nearest-neighbour sampling yields sea (1), coastal land (3), or inland (4). Omitting it explicitly selects inland zone 4 and is inappropriate for rigorous coastal predictions. A complete coastal implementation must also compute the terminal-to-coast path distances required by P.1812.

3.3 Rasterization and encoding

The polar matrix is bilinearly interpolated in azimuth and radius to a default 60 m AEQD output grid. This grid spacing is a publication choice, not a statement that the underlying propagation model resolves every 60 m feature. The raster is then reprojected to EPSG:4326.

Numerical PNG tiles use an unsigned 8-bit code q. Zero is NoData, and

$$q = 1 + \mathrm{round}\left(\frac{E - E_{\min}}{\Delta E}\right), \qquad E_{\min} = -20, \qquad \Delta E = 0.5 \text{ dB}, \tag{3}$$

with saturation to $[1, 255]$. Presentation tiles apply a fixed absolute scale from purple through blue, cyan, green, yellow and red. A threshold from 0 to 50 dB μV m^{-1}, default 5, changes alpha only. It never changes stored predictions. Station-independent colours make overlapping repeaters comparable.

4 Expected S-meter indication

The primary output remains E in dB μV m^{-1}. For an ideal polarization-matched, lossless, 0 dBisotropic receiving antenna terminated in a matched receiver, plane-wave power density and effective aperture yield

$$P_r\,[\mathrm{dBm}] = E\,[\mathrm{dB}\mu\mathrm{V/m}] + 20\log_{10}\lambda\,[\mathrm{m}] - 126.755. \tag{4}$$

This is consistent with the free-space transmission framework introduced by Friis [4]. The nominal IARU Region 1 recommendation assigns S9 to −73 dBm below 30 MHz and −93 dBm above it, with 6 dB per S unit below S9 [5]. Since P.1812 begins at 30 MHz, the latter reference normally applies here.

Equation 4 does not include real antenna gain and direction, height-dependent local coupling, body loss, polarization mismatch, cable loss, receiver bandwidth, AGC calibration, or building penetration. The interface therefore says *expected S-meter indication*, presents the corresponding field value, and labels the idealized receiver reference.

5 Implementation and M5 optimization

P.1812 endpoint calculations are branch-heavy and largely independent by azimuth. On native Apple silicon, the engine queries the performance-core count and reserves one or two such cores for the operating system. A process pool distributes ray chunks. Numeric-library thread counts are limited to one per worker, avoiding multiplicative oversubscription.

Table 1: Portable acceleration roadmap.

Target	Suggested implementation and evidence
Linux	NUMA-aware pools, shared memory, SIMD profiling
Windows	Spawn-safe mappings and Job Object resource limits
NVIDIA	CUDA/CuPy interpolation and colour kernels
AMD	ROCm/HIP implementation of the same kernel contract
Intel	oneAPI/SYCL on integrated and discrete GPUs
Cross-vendor	Vulkan compute; WebGPU for client composition

macOS uses the safe `spawn` start method after GDAL/NumPy initialization. Naively passing terrain arrays would copy them into every process. The M5 path places terrain and climatic-zone grids in POSIX shared memory; workers attach read-only NumPy views. This reduces memory pressure and serialization time.

The polar-to-Cartesian step is regular and massively data-parallel. An optional MLX backend launches one fused Metal kernel that computes geometry, indices, four samples, bilinear weights, and the output value. The P.1812 model itself stays on CPU performance cores. This division matches workload structure and keeps a NumPy reference backend for numerical comparison. Docker Desktop Linux guests cannot access the host Metal device, so GPU computation requires a native macOS process.

Every accelerator must be compared with the NumPy backend using maximum absolute and relative error, and must report hardware, OS, runtime, wall time, peak memory, and energy where available. Moving the whole P.1812 algorithm to a GPU is not automatically beneficial because divergence and many short profiles can negate throughput advantages.

6 API and reproducibility contract

Each station artifact contains a floating-point GeoTIFF and a JSON manifest. The manifest records model identifier, pinned implementation, algorithm version, physical units, coordinates, frequency, ERP, antenna height, polarization, radius, profile/radial/output spacing, ray count, worker count, raster backend, field range, and assumptions. Input rasters should additionally be identified by checksums, vertical datum and license in production workflows.

A stateless read-only HTTP service exposes the manifest, thresholded colour tiles, immutable numerical tiles, and point samples. The service reads artifact files rather than exposing an operational database to browsers. This separation supports TLS termination, caching, rate limiting, atomic artifact replacement, and independent calculator credentials.

The TypeScript client keeps one imagery layer per station. Filter changes map to the exact selected station set. Layer toggles and base-map changes do not invoke camera movement. Pointer sampling is debounced and cancellable; when several visible stations intersect, the strongest modeled field is shown.

7 Validation

Three validation levels must remain distinct.

Analytical and encoding tests. P.525 reference values, field/power conversion, S thresholds, tile geometry, NoData, quantization round trips, and azimuth wraparound are deterministic unit tests.

Implementation conformance. The pinned Py1812 validation profiles should reproduce the recommendation's reference outputs within their specified tolerance. Synthetic flat, ridge, and coastal profiles exercise wrapper logic. NumPy and Metal rasterizers must agree within declared floating-point tolerances. Conformance demonstrates consistent implementation, not predictive truth at a location.

Empirical validation. Calibrated field surveys require known antenna factor, receiver bandwidth, position, time, transmitter ERP, antenna pattern, cable loss, and preferably pre-registered routes. Calibration and validation routes should be disjoint. Report bias, mean absolute error, root-mean-square error, standard deviation and residual quantiles by distance, terrain class, visibility state and field band. Uncalibrated consumer S meters are useful qualitative observations but are not absolute field instruments.

8 Comparison with binary visibility maps

Compared with a single-colour terrain-visibility footprint, the specified method offers six concrete advantages:

1. the map retains a physical quantity and absolute units;
2. frequency, ERP, heights, polarization, time and location enter the model;
3. diffraction and longer-range P.1812 mechanisms extend beyond geometry;
4. thresholds can change without recalculating or recolouring the physics;
5. numerical tiles support composition and point interrogation; and
6. manifests make assumptions and revisions auditable.

These benefits do not remove uncertainty. A simple visibility map may be faster, easier to explain, and sufficient for preliminary siting. The appropriate claim is improved information content and physical traceability, not guaranteed local reception or universal superiority.

9 Limitations and future work

Current limitations include isotropic transmitter patterns, zero default clutter, a median-statistics default, no indoor model, and possible DEM surface bias. Coastal predictions need verified zone and coast-distance preprocessing. Future work should add antenna-pattern rotation, provenance-rich clutter, uncertainty bands, field-survey tooling, residual dashboards, and sensitivity analysis across DEMs and statistical percentages.

Software acceleration should expand through tested CUDA, ROCm, SYCL and Vulkan backends. WebGPU can combine multiple numerical tiles locally and reduce API load. Such work must preserve the field-value contract and publish numerical equivalence tests.

10 Conclusion

Terrain-aware repeater mapping is most useful when it preserves what was calculated. The method presented here produces continuous P.1812 field strength in dB µV m^{-1}, uses an explicit P.525 near zone, separates numerical values from presentation colours, and makes any S-meter indication conditional on stated receiving assumptions. Provenance manifests and layered validation prevent a visually attractive overlay from being mistaken for measured truth. Apple M5 optimizations reduce runtime without changing the scientific contract, while the reference NumPy path and documented porting interfaces support independent implementations.

Software and data availability

The source code, web client, API, tests, LaTeX manuscript and citation metadata are released under the MIT License. Py1812, ITU digital products, elevation data and map imagery retain their separate licenses and are not redistributed. A versioned archive should be cited for scientific use.

Acknowledgements

The author acknowledges the ITU-R Study Group 3 propagation work, the Py1812 contributors, the IARU technical recommendations, and the providers and validators of global elevation data.

References

[1] Airbus Defence and Space. Copernicus digital elevation model: Product handbook. Technical Report GEO1988-COPERNICUSDEM-SPE-002, Issue 5.0, European Space Agency and European Union, November 2022. URL: https://dataspace.copernicus.eu/explore-data/data-collections/copernicus-contributing-missions/collections-description/COP-DEM, doi:10.5270/ESA-c5d3d65.

[2] Kenneth Bullington. Radio propagation at frequencies above 30 megacycles. *Proceedings of the IRE*, 35(10):1122–1136, 1947. doi:10.1109/JRPROC.1947.232600.

[3] Jacques Deygout. Multiple knife-edge diffraction of microwaves. *IEEE Transactions on Antennas and Propagation*, 14(4):480–489, 1966. doi:10.1109/TAP.1966.1138719.

[4] Harald T. Friis. A note on a simple transmission formula. *Proceedings of the IRE*, 34(5):254–256, 1946. doi:10.1109/JRPROC.1946.234568.

[5] International Amateur Radio Union Region 1. Hf managers' handbook, s-meter recommendation. Technical report, IARU Region 1, 2016. URL: https://www.iaru-r1.org/wp-content/uploads/2019/12/IARURegion1HFManagerHandbook8.2.1.pdf.

[6] International Telecommunication Union. Calculation of free-space attenuation. Technical Report Recommendation ITU-R P.525-5, ITU Radiocommunication Sector, November 2024. URL: https://www.itu.int/rec/R-REC-P.525-5-202411-I/en.

[7] International Telecommunication Union. A path-specific propagation prediction method for point-to-area terrestrial services in the frequency range 30 mhz to 6 ghz. Technical Report Recommendation ITU-R P.1812-8, ITU Radiocommunication Sector, September 2025. URL: https://www.itu.int/rec/R-REC-P.1812-8-202509-I/en.

[8] Jana Marešová, Kateřina Gdulová, Petra Pracná, David Moravec, Lukáš Gábor, Jan Prošek, Vojtěch Barták, and Vítězslav Moudrý. Applicability of data acquisition characteristics to the identification of local artefacts in global digital elevation models: Comparison of the copernicus and tandem-x dems. *Remote Sensing*, 13(19):3931, 2021. doi:10.3390/rs13193931.

[9] Evgenia Vitanova et al. Py1812: Python implementation of recommendation itu-r p.1812. GitHub repository, revision a5205e6a65db27391a8ba79bd5a365e5391f9fdf, 2025. URL: https://github.com/eeveetza/Py1812.